Matheus Coutinho da Silva

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EDUCATION

Amherst College, Amherst, MA, USA — Aug. 2021 – May 2025
Computer Science, Bachelor of Arts (*cum laude*), GPA: 3.84/4.00

SELECTED EXHIBITIONS

Group Exhibition, *Oh, My Dear*, curated by Professor Robert T. Sweeney — **August 2026**
Fayerweather Hall, Amherst College, Amherst, MA, USA

Group Exhibition, *I Saw You First*, curated by Professor Gabriel Phipps — **July 2026**
Fayerweather Hall, Amherst College, Amherst, MA, USA

Group Exhibition, *The End of The World*, curated by Professor Gabriel Phipps — **April 2026**
Fayerweather Hall, Amherst College, Amherst, MA, USA

Group Exhibition, *Restless*, curated by Professor David I. Gloman — **March 2026**
Fayerweather Hall, Amherst College, Amherst, MA, USA

Group Exhibition, *Quiet!*, curated by Professor Robert T. Sweeney — **February 2026**
Fayerweather Hall, Amherst College, Amherst, MA, USA

Group Exhibition, *Within*, curated by Professor Robert T. Sweeney — **January 2026**
Fayerweather Hall, Amherst College, Amherst, MA, USA

Group Exhibition, *Do Not Trust*, curated by Professor David I. Gloman — **December 2025**
Fayerweather Hall, Amherst College, Amherst, MA, USA

SELECTED FEATURES

Featured Artist, BRASA (Brazilian Student Association), Online, Brazil — Jun. 2026 – Oct. 2026

Animation, selected creative coding works — **October 2026**
Interactive Art, selected creative coding works — **September 2026**
Environment Design, selected 3D works — **August 2026**
Symbolic Art, selected 3D works — **July 2026**
Horror, selected 3D works — **June 2026**

PROFESSIONAL AND CREATIVE EXPERIENCE

Torus Knot 3D Designs, Holyoke, MA, USA — Jun. 2025 – Present
Architectural & 3D Designer

- Delivered architectural visualizations, including 115+ renderings and 2 animated walkthroughs totaling 15+ minutes.
- Designed 20+ custom furniture, jewelry, and sculptural models, including a life-size table and a 3-foot cockerel sculpture; developed 35+ fabrication-ready models and 15+ physical prototypes.
- Prepared 60+ CAD drawing sheets for commercial glazing and curtain wall systems, translating architectural specifications into fabrication and installation documentation.

iD Tech Camps, Manchester, NH, USA Jun. 2025 – Aug. 2025
On-Campus Instructor: Machine Learning & Roblox Development

- Instructed 12 students across 2 tracks, teaching Roblox development to 5 students (ages 7–9) and Python, AI, and machine learning to 7 students (ages 13–17).
- Guided students in building machine learning models and AI-powered assistants using NumPy, Teachable Machine, and OpenAI tools, as well as playable Roblox games using Roblox Studio.
- Achieved a 100% student project completion rate, with every student completing an AI project or publishing a playable game; recognized by camp director for classroom leadership.

RESEARCH & SOFTWARE ENGINEERING

Amherst College, Amherst, MA, USA Sep. 2024 – May 2025
Honors Thesis Researcher - Conversational AI System

- Designed and implemented a conversational AI agent integrating 3 core capabilities: personality, emotion, and long-term memory to support adaptive, human-like interactions.
- Developed 8 modular components for goal-driven dialogue, emotional response modeling, memory retrieval, and context-aware conversations.
- Conducted 100+ experimental conversations to evaluate memory retention, personality consistency, and response quality, refining the agent through iterative testing.

Georg-August-Universität Göttingen, Göttingen, Lower Saxony, Germany Mar. 2024 – Aug. 2024
Student Researcher - Deep Learning for Natural Language Processing

- Developed and trained BERT- and BART-based models across 5 NLP tasks, including sentiment analysis, semantic similarity, and paraphrase generation.
- Optimized distributed training workflows across multi-GPU clusters, reducing model training time by over 40%.
- Improved Quora Question Pairs accuracy from 75.9% to 88.0% through hyperparameter tuning and cross-validation, earning 1st place in the task competition.

University of Massachusetts Amherst, Amherst, MA, USA Jun. 2023 – Jul. 2023
Research Assistant Intern - Word Meaning Modeling

- Developed and trained 3+ neural network models to investigate how computers associate word structures with their meanings.
- Conducted 50+ experiments using HPC clusters and Google Colab, evaluating model performance and optimizing training workflows.
- Built 2+ visualization and diagnostic tools and left written documentation to train current/future research team members in computational workflows and research tools.

Amherst College, Amherst, MA, USA Feb. 2022 – Dec. 2022
Java Developer

- Developed a Java Swing application featuring 4 interactive puzzle games, with graphical interfaces, instructions, and navigation.
- Built a Java lottery simulation supporting 2-, 3-, and 4-digit betting combinations, with randomized drawings, automated payouts, and player balance management.
- Applied object-oriented programming principles across both projects to organize game logic, implement interactive features, and develop reusable, maintainable code.

MENTORING AND LEADERSHIP EXPERIENCE

AI in The Liberal Arts Initiative (Amherst College), Amherst, MA, USA Feb. 2023 – Dec. 2023
AI Tools and Mentorship Intern, Supervisor: Professor Lee Spector

- Led 5+ campus discussions and demonstrations connecting AI research with applications in the arts and humanities.
- Developed 8+ tutorials covering AI applications in literary analysis, historical data analysis, and AI-generated multimedia.
- Mentored 25+ students and faculty in using AI and machine learning tools for academic research and creative projects.

SCHOLARSHIPS, CERTIFICATIONS & AWARDS

B.A. awarded *cum laude*. Amherst College, 2025.

Associate Member Nominee. Sigma Xi - The Scientific Research Honor Society, 2025.

1st Place (Individual). Quora Question Pairs Task, Deep Learning Systems Final Competition (Georg-August-Universität Göttingen), 2024.

1st Place (Team). Overall Team, Deep Learning Systems Final Competition (Georg-August-Universität Göttingen), 2024.

Scholarship Recipient. Brazilian Student Association (BRASA) [12 recipients out of 287 applicants], 2021.

Fellow. Prep Estudar Fora. Selective Fundação Estudar program that prepares Brazilian students for admission to universities abroad [46 admitted out of 7561 applicants], 2020.

Finalist. EF Challenge 2020. International public speaking competition focused on global issues [8 finalists selected out of 1,150 participants], 2020.

Youth Ambassador. Youth Ambassadors Program, USA. U.S. Department of State exchange program focused on leadership, civic engagement, and cross-cultural exchange [50 admitted out of 7,000+ applicants], 2020.

Silver medal. Brazilian Biology Olympiad (OBB) [only silver medalist from Rio de Janeiro; 47 silver medals out of 125,000 contestants], 2019.

Fellow. Clubes de Ciência Brasil (Science Clubs Brazil). Science immersion and mentorship program in which students develop research projects with university researchers [100 admitted out of 500 applicants], 2019.

Golden Feather Student. Awarded to the valedictorian of Colégio Pedro II [1 out of 255 students], 2019.

Fellow. Latin American Leadership Academy, Ecuador [26 admitted out of 700 applicants], 2019.

Finalist. Prêmio Prudential Espírito Comunitário. Brazilian youth volunteerism award recognizing community service initiatives with social impact [10 finalists out of 1,500+ applicants], 2019.

SELECTED SKILLS

Programming & Scientific Computing: Python, Java, JavaScript, PHP, MATLAB
Web Development: HTML, CSS, Bootstrap
Design & Simulation Tools: AutoCAD, Revit, SketchUp, Blender, Roblox Studio, Adobe Acrobat, Processing
Development & Tools: Git (GitHub), VS Code, Overleaf
Data & Analytics: Pandas, scikit-learn, TensorFlow, PyTorch, Transformers, SentenceTransformers, Excel
Other Competencies: Google Colab, HPC clusters
Languages: Portuguese (native), English (fluent), Spanish (intermediate), French (basic), German (basic)